Interpretable Regression Baselines and Diagnostics

SARCOS robot arm inverse dynamics | DS-A3

Research question: how accurately do linear baselines predict first-joint torque from 21 kinematic features, and how does error vary across observed states?

Motivation: inverse dynamics maps an observed robot state to torque demand, a meaningful regression target for studying predictive baselines relevant to robot control and motion planning. Interpretable baselines help establish performance and error boundaries before adding complexity; this study evaluates prediction, not control effectiveness.

Dataset card: the 44,484 × 28 numeric source matrix contains 21 kinematic inputs (seven positions, seven velocities, seven accelerations) and seven joint torque outputs. The target is first-joint torque (column 21, zero-based); paired continuous state and torque observations suit supervised regression. The pinned public Kaixhin/SARCOS GitHub revision is the download source; GPML documents the dataset and credits Sethu Vijayakumar.

Provenance and scope: 4,446 of 4,449 aligned rows in the historical 4,449 × 28 test matrix match the source exactly, and all are within 4.0000001e-6 maximum cell difference. It was excluded from modeling and evaluation. Rows are dependent, reliable episode identifiers are unavailable, and the 25-row purge cannot establish independent trajectories. Physical units and dataset redistribution terms remain unverified. Dataset SHA256: b8a249733253ba6097372fedee7696833fcf30de42037d5b4a7227f21a6d1d97

Method: training-mean Dummy, standardized OLS, and standardized Ridge. Scaling is fit within each training fold. Ridge alpha is selected by validation MAE from the preregistered grid; final models are refit on Train + Validation and evaluated once on Test.

Blocked uses contiguous source-row regions with the frozen 25-row boundary purge. Random uses frozen random partitions. Neither establishes episode-independent generalization.

Reproduction: activate nussem1, run the notebook, then run scripts/validate_project.py and python -m unittest discover -s tests -v. Environment: Python: 3.10.18 | NumPy: 2.2.6 | pandas: 2.3.1 | SciPy: 1.15.2 | scikit-learn: 1.7.1 | Platform: macOS-15.5-arm64-arm-64bit

Predictive performance

MAE is the primary metric; RMSE emphasizes larger residuals and R^2 summarizes variance explained. H1 is supported on the tested blocked evaluation: OLS materially improves on the training-mean baseline.
Ridge provides no meaningful improvement in this tested setting. This conclusion is limited to these models, alpha grid, dataset, and evaluation protocols.

Protocol	Partition	Model	MAE	RMSE	R^2
Blocked	Validation	Dummy mean	14.1541	18.9393	-0.0142
Blocked	Validation	OLS	3.6591	5.1754	0.9243
Blocked	Validation	Ridge	3.6591	5.1754	0.9243
Blocked	Test	Dummy mean	11.6312	15.0777	-0.0076
Blocked	Test	OLS	3.8650	5.3396	0.8736
Blocked	Test	Ridge	3.8650	5.3396	0.8736
Random	Validation	Dummy mean	15.3280	20.3378	-0.0004
Random	Validation	OLS	3.9461	5.5460	0.9256
Random	Validation	Ridge	3.9461	5.5460	0.9256
Random	Test	Dummy mean	15.3064	20.2932	-0.0000
Random	Test	OLS	3.9471	5.5700	0.9247
Random	Test	Ridge	3.9471	5.5700	0.9247

Evaluation design and Ridge selection

The blocked protocol holds out contiguous row regions, with the manifest-defined purge gaps; it is sensitive to source ordering and cannot verify episode independence. The random protocol samples held-out rows from the mixed source distribution and may leave nearby observations across partitions.

Blocked validation has markedly lower velocity and acceleration norms than its training and test regions, indicating distribution shift. Random partitions have more similar summaries. The protocol comparison is descriptive and does not by itself establish leakage.

Alpha was selected using Validation MAE only. Test labels did not enter preprocessing, fitting, hyperparameter selection, slice thresholds, or model decisions.

Protocol	Partition	n	Target mean	Median velocity norm	Median acceleration norm
Blocked	Train	26,665	13.3676	2.3018	21.9614
Blocked	Val	8,847	15.6105	0.3636	5.4212
Blocked	Test	8,872	12.6192	2.2242	24.6873
Random	Train	26,690	13.5860	2.0344	19.6477
Random	Val	8,897	13.9716	2.0266	19.4187
Random	Test	8,897	13.5394	2.0118	19.4866

Alpha	Blocked validation MAE	Random validation MAE
0.0010	3.6591	3.9461
0.0100	3.6591	3.9461
0.1000	3.6593	3.9462
1.0000	3.6611	3.9462
10.0000	3.6793	3.9472
100.0000	3.8384	3.9619
1000.0000	4.6976	4.2184

Protocol	Selected alpha
Blocked	0.0010
Random	0.0010

Descriptive error slices

Slices use Blocked Train quartile cutoffs, applied unchanged to Blocked Test. Open-ended end labels preserve the actual digitize assignments; the outside-range count indicates test observations beyond the training min/max.

Higher acceleration and higher absolute torque slices have higher absolute errors. Velocity-slice error is non-monotonic: the low-velocity group has higher MAE than the highest-velocity group. These are descriptive associations. Higher absolute error at large torque does not establish higher relative error or a physical causal mechanism.

Slice variable	Slice interval (train-derived cutoffs)	n	Outside train range count	MAE	RMSE
Velocity norm	(-inf, 1.44069]	2,552	0	4.6970	6.6465
Velocity norm	(1.44069, 2.30176]	2,110	0	3.5896	4.9702
Velocity norm	(2.30176, 2.99481]	2,366	0	3.4456	4.5172
Velocity norm	(2.99481, +inf)	1,844	0	3.5666	4.6465
Acceleration norm	(-inf, 13.8308]	1,922	0	3.1327	4.0621
Acceleration norm	(13.8308, 21.9614]	1,891	0	3.5183	4.6057
Acceleration norm	(21.9614, 31.1808]	2,624	0	4.0954	5.4923
Acceleration norm	(31.1808, +inf)	2,435	0	4.4638	6.4713
Absolute torque	(-inf, 7.11475]	2,210	0	3.2920	4.2341
Absolute torque	(7.11475, 15.3596]	2,541	0	3.1628	4.2628
Absolute torque	(15.3596, 28.7719]	2,779	0	3.8462	5.1743
Absolute torque	(28.7719, +inf)	1,342	0	6.1770	8.3137

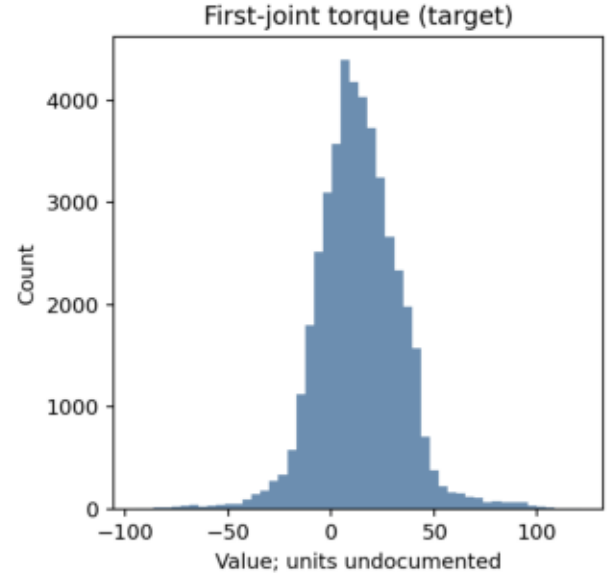
Coefficient summary

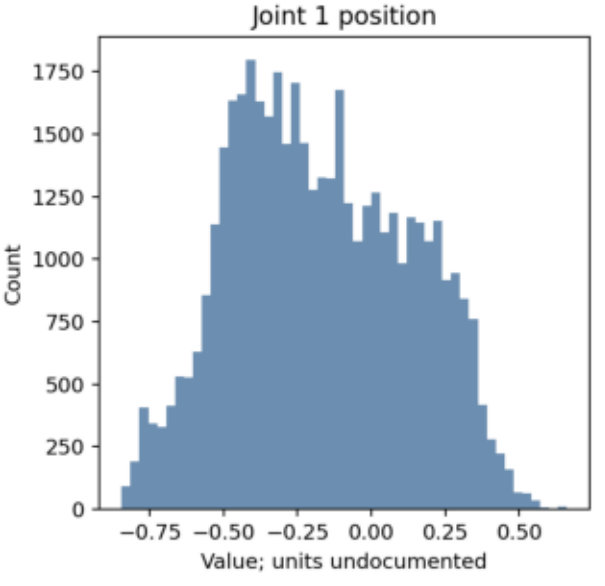
The displayed standardized coefficients are descriptive. Correlated kinematic features limit coefficient-level interpretation; they should not be read as separately identified physical parameters.

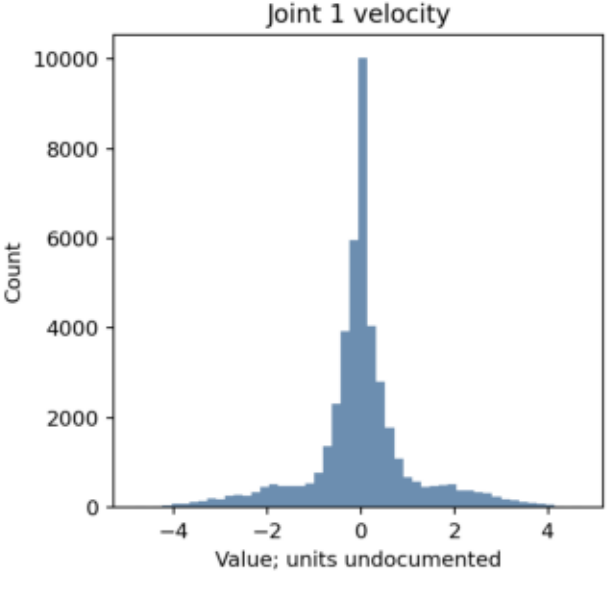
Feature	OLS coef.	Ridge coef.
joint_1_acceleration	26.1780	26.1780
joint_4_acceleration	9.7529	9.7529
joint_1_position	8.7555	8.7555
joint_4_position	-2.7748	-2.7748
joint_5_position	2.0401	2.0401
joint_6_position	1.9672	1.9672
joint_3_acceleration	-1.9135	-1.9135
joint_2_acceleration	-1.7466	-1.7466
joint_5_acceleration	-1.3694	-1.3694
joint_3_position	-1.1819	-1.1819
joint_2_position	0.9444	0.9444
joint_7_position	-0.7501	-0.7501

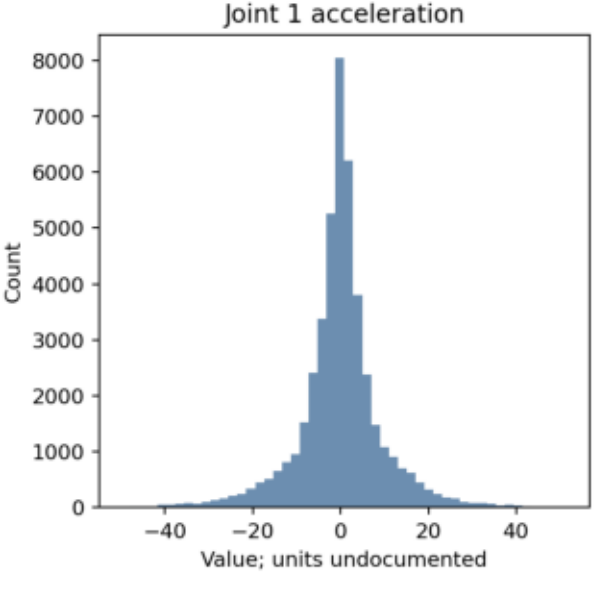
Data exploration

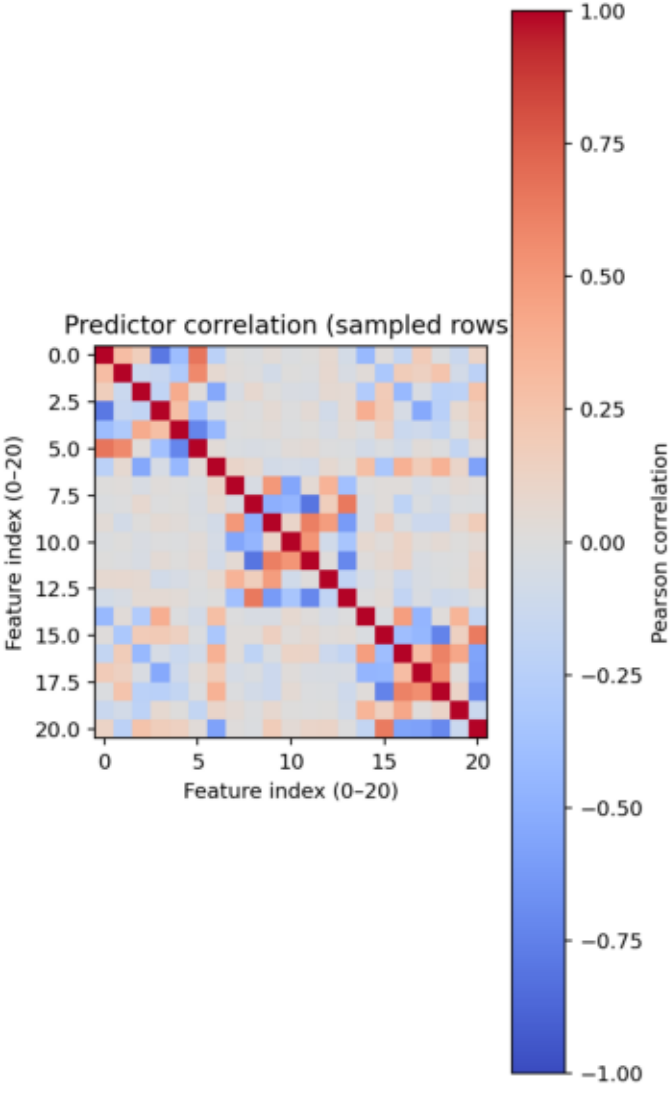
Training data summaries show broad operating-state variation. Marginal distributions do not establish temporal independence or identify physical units.





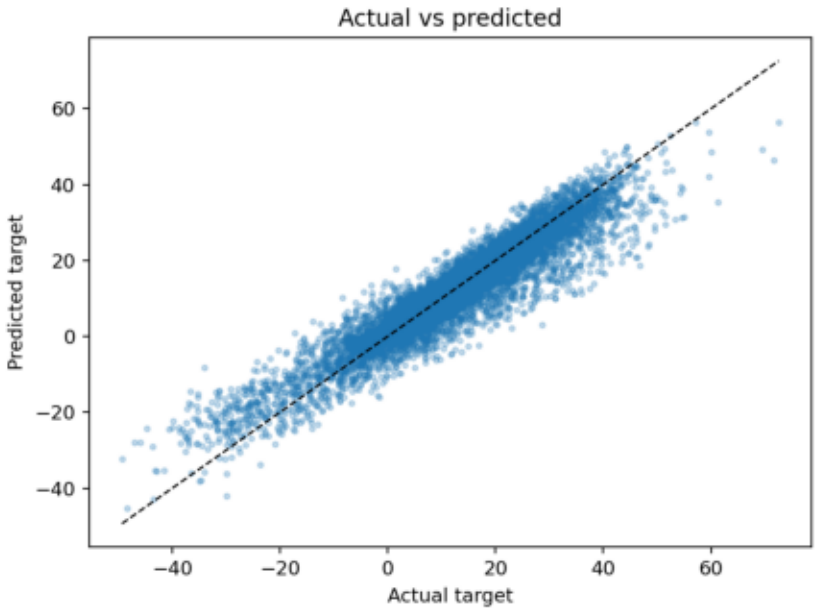


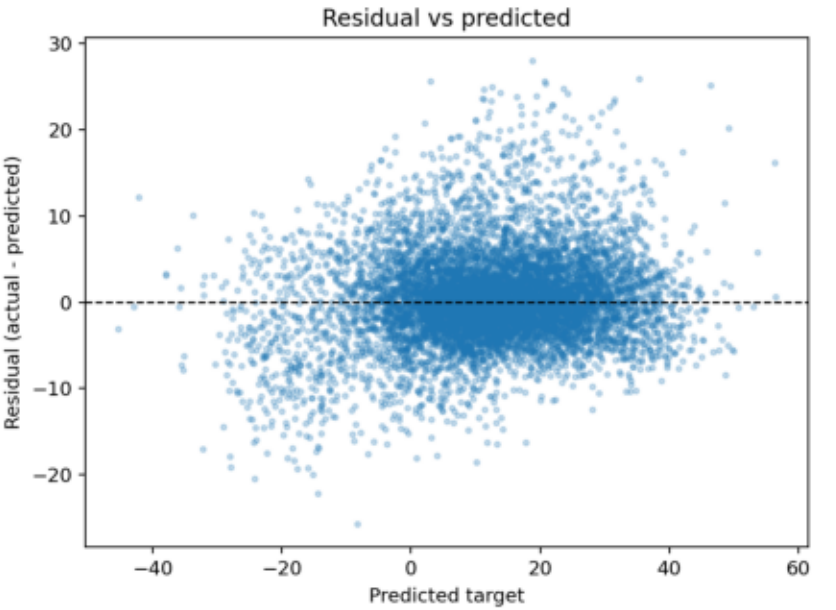


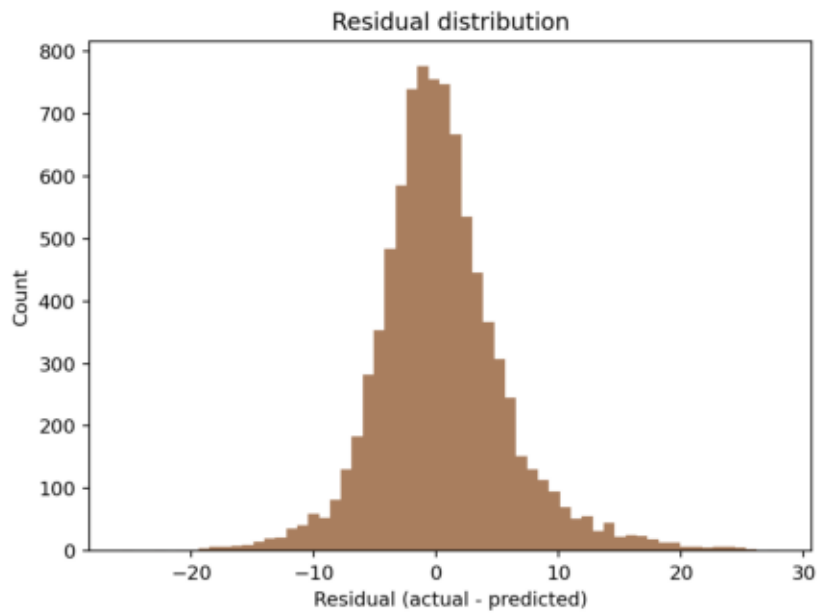


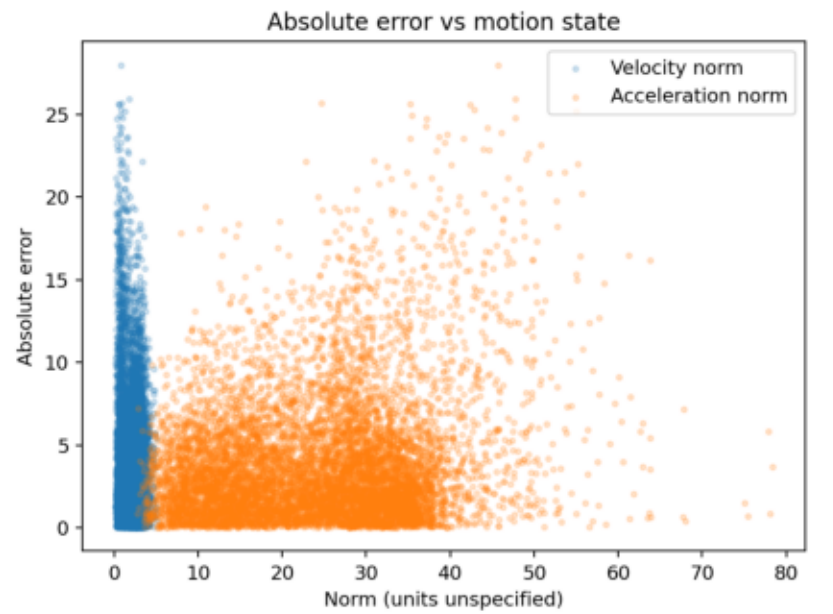
Blocked test residual diagnostics

Residual plots characterize the final Blocked Ridge fit on held-out rows. They are diagnostic descriptions, not evidence of episode-level transfer.



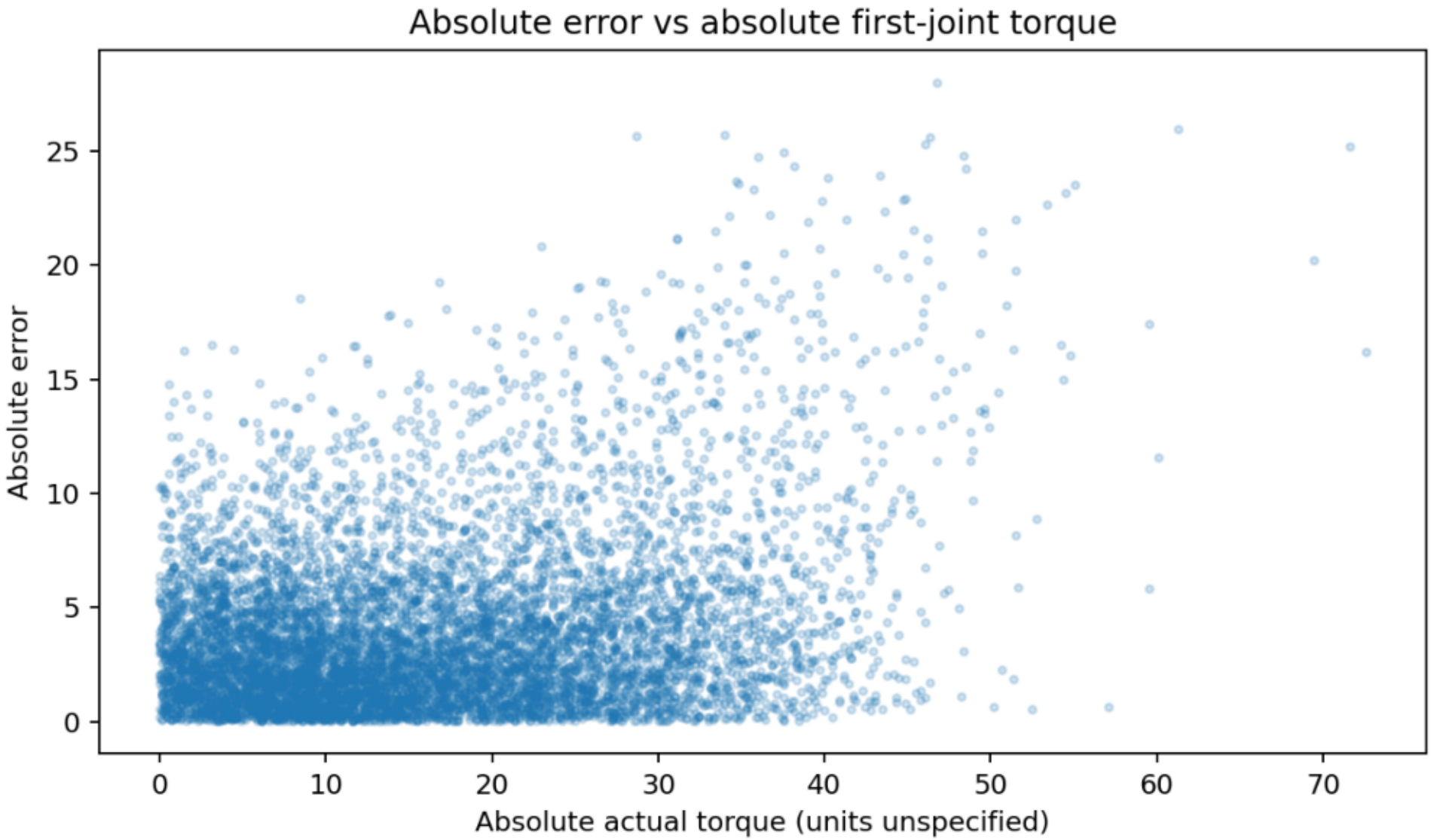






Absolute error versus absolute torque

The spread in absolute error increases in parts of the observed torque range. This plot does not normalize error by torque, so it cannot support claims about relative error.



Additional Blocked OLS diagnostics

Influence statistics are computed on Blocked Train using an OLS model fit to those rows. Leverage is the diagonal of the projection matrix computed from a reduced QR factorization; no full n-by-n matrix is formed. Cook's distance uses the standard squared-residual/leverage expression.

Heuristic flags use leverage > 2p/n and Cook's distance > 4/n. These thresholds identify observations for inspection; they do not establish data errors. The Breusch–Pagan auxiliary regression uses the 21 standardized predictors and training-fit squared residuals. Its null is constant conditional error variance. The conventional chi-square calibration assumes independent, correctly specified errors; strong serial dependence can make its p-value unreliable.

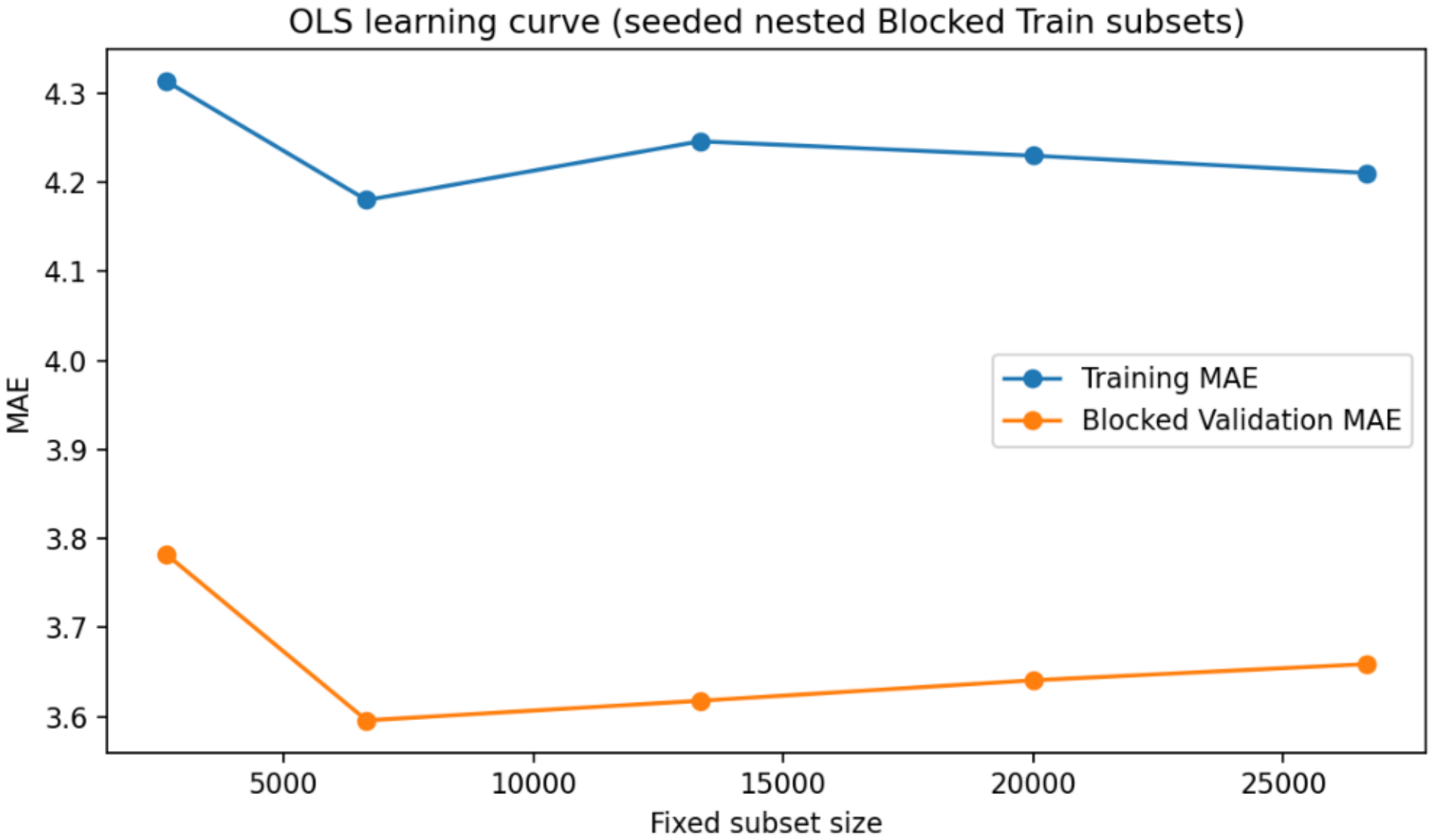
Compare the test residual-versus-fitted plot with this training-only test. Both are descriptive; the test statistic is not used to change models.

BP LM statistic	BP degrees freedom	BP p-value	High leverage rows
1622.3322	21.0000	0.0000	1,891

Original row	Leverage	Studentized residual	Cook distance	Above 4/n Cook cutoff
24,152	0.0030	-4.7593	0.0031	1.0000
25,355	0.0025	-4.7320	0.0026	1.0000
3,210	0.0009	7.6505	0.0025	1.0000
24,153	0.0031	-4.1337	0.0024	1.0000
25,354	0.0026	-4.3330	0.0022	1.0000

OLS learning curve: Blocked Train and Validation

Each fixed subset is a nested sample from a seeded permutation of frozen Blocked Train rows. Training MAE is measured on the fitted subset; Validation MAE is measured on the same frozen Blocked Validation set at every size. The curve is descriptive under this sampling design, especially because the Blocked Validation region has a distinct low-motion distribution.



Post-hoc Blocked Test error cases

These are the five largest absolute residuals from the already evaluated Blocked Ridge model, ranked after the original evaluation. They are shown for descriptive inspection only and were not used to tune or select any model. Torque magnitude is outcome-conditioned; velocity and acceleration norms are input-derived.

Large residuals identify cases where this fitted model performed poorly in this held-out segment. The table does not establish a physical cause or imply that the observations are erroneous.

Original row	Actual	Prediction	Residual	Absolute residua	Velocity norm	Acceleration norm	Absolute target
39,383	46.7586	18.7520	28.0066	28.0066	0.8530	45.6971	46.7586
43,519	61.3027	35.3512	25.9515	25.9515	1.7395	47.7508	61.3027
35,811	-33.9815	-8.2962	-25.6853	25.6853	0.9109	24.7076	33.9815
43,988	28.6804	3.0457	25.6347	25.6347	0.5369	35.2212	28.6804
35,690	46.3508	20.7463	25.6045	25.6045	0.6503	42.3352	46.3508

Exploratory Train/Validation model comparison

Lasso and HistGradientBoostingRegressor are post-hoc exploratory candidates. Their alpha/configuration comparisons use only Blocked Train and Blocked Validation; the previously inspected Test was not used for their selection or tuning. Do not interpret their validation scores as confirmatory estimates.

Lasso zero coefficients are counted using the documented absolute tolerance. The fixed small HistGradientBoosting search provides one modest nonlinear comparison; its added flexibility costs interpretability and computation. Validation improvements, if any, are specific to this shifted validation region.

Model	Selection	Alpha/config	Validation MAE	Zero coefficients (tol=1e-8)
Dummy mean	Fixed baseline	—	14.1541	—
OLS	Fixed baseline	—	3.6591	—
Ridge	Fixed validation selection	0.001	3.6591	—
Lasso	Validation MAE over fixed alpha	0.001	3.6591	0
HistGradientBoostingRegressor	Validation MAE over fixed, modest configuration	{'learning_rate': 0.05, 'max_iter': 120, 'max_leaf_nodes	5.0114	Not applicable

Negative results and claim boundary

Ridge's improvement over OLS was negligible in the original tested setting. The exploratory additions are validation-only and do not replace or revise the original reported Test results.

The near-complete overlap in the supplied historical SARCOS files is a data-provenance warning, not a usable independent benchmark: 4446/4449 rows are exactly equal to their aligned source row; all 4449/4449 are within 4.0000001e-6 maximum absolute cell difference (observed maximum 4e-06).

The project establishes predictive behavior only for these models, data, and frozen protocols. The Blocked Validation segment is markedly low-motion and unrepresentative of the other blocked partitions; this limits its usefulness for broader tuning. We therefore did not add blocked cross-validation or redesign tuning under the deadline. Serial dependence, unknown episode identifiers, and the local purge gap remain limitations.

Standardized regression coefficients summarize conditional linear associations under this fitted feature set, not physical parameters or causal effects. Correlated kinematic predictors make individual coefficients unstable to feature dependence. Inverse-dynamics prediction estimates torque from observed state; it does not estimate the effect of intervening on a joint state while controlling the physical system and confounders.

Discussion, limitations, and conclusion

Research question answered within scope: the tested first-joint torque can be predicted substantially better than the training-mean baseline on the frozen blocked holdout (OLS MAE 3.865 versus 11.631). This is evidence for predictive capability on these rows, not a guarantee for deployment or another trajectory.

Complexity result: Ridge does not meaningfully improve on OLS under the tested models, alpha grid, dataset, and protocols. Validation-only Lasso is essentially comparable to OLS in the tested grid, while the small HistGradientBoosting search has higher Blocked Validation MAE; these post-hoc comparisons are exploratory and have no new Test evaluation.

Reliability findings: higher acceleration and higher absolute-torque slices have higher absolute errors; velocity-slice error is non-monotonic. Residual plots show remaining structure and changing error spread. The training-only Breusch–Pagan statistic is large (1622.33, 21 df), but serial dependence weakens conventional chi-square p-value calibration. Leverage/Cook cutoffs flag observations for inspection only; they do not establish invalid data or causal influence. The learning curve's Validation MAE is lowest at the 25% subset and then changes modestly, a descriptive pattern under this fixed subset and shifted validation region.

Error slices and the five largest post-hoc Test residuals locate observed failure regions but do not explain their physical cause. The absolute-torque slice conditions on the outcome; high absolute error at high torque does not imply high relative error. Standardized coefficients are conditional predictive associations, not physical parameters or causal effects.

Limitations: source-row dependence is substantial; timestamps and reliable episode identifiers are unavailable; the local 25-row purge does not establish independence. Blocked regions show distribution shift, and random partitions may retain nearby observations. The original test matrix overlaps the training matrix and is not an independent benchmark. No claim is made about new robots, tasks, or episodes.

The target's physical units and the dataset's redistribution license remain unverified. The public GitHub repository is the verified download source; the upstream GPML page is the dataset description and attribution reference. Further use or redistribution should follow the dataset owner's terms once verified.

References: Kaixhin/SARCOS, https://github.com/Kaixhin/SARCOS ; GPML dataset page, https://gaussianprocess.org/gpml/data/ ; split manifest: data/splits/sarcos_split_manifest.json ; protocol: docs/experiment_protocol.md.